

PHYSICS  
CLASS

$$E = m \cdot c^2$$

$$P = \frac{F}{A}$$

$$V = a \cdot t$$

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$



PHYSICS II (Lecture 5) Dynamics Physics I  
By Dr. Rabee M. Hagem and Dr. Ahmed Samir

## LEARNING OBJECTIVES

### Theory

- Describe the meaning of Dynamics Physics
- Explain how an object moves in terms of the forces that change its motion

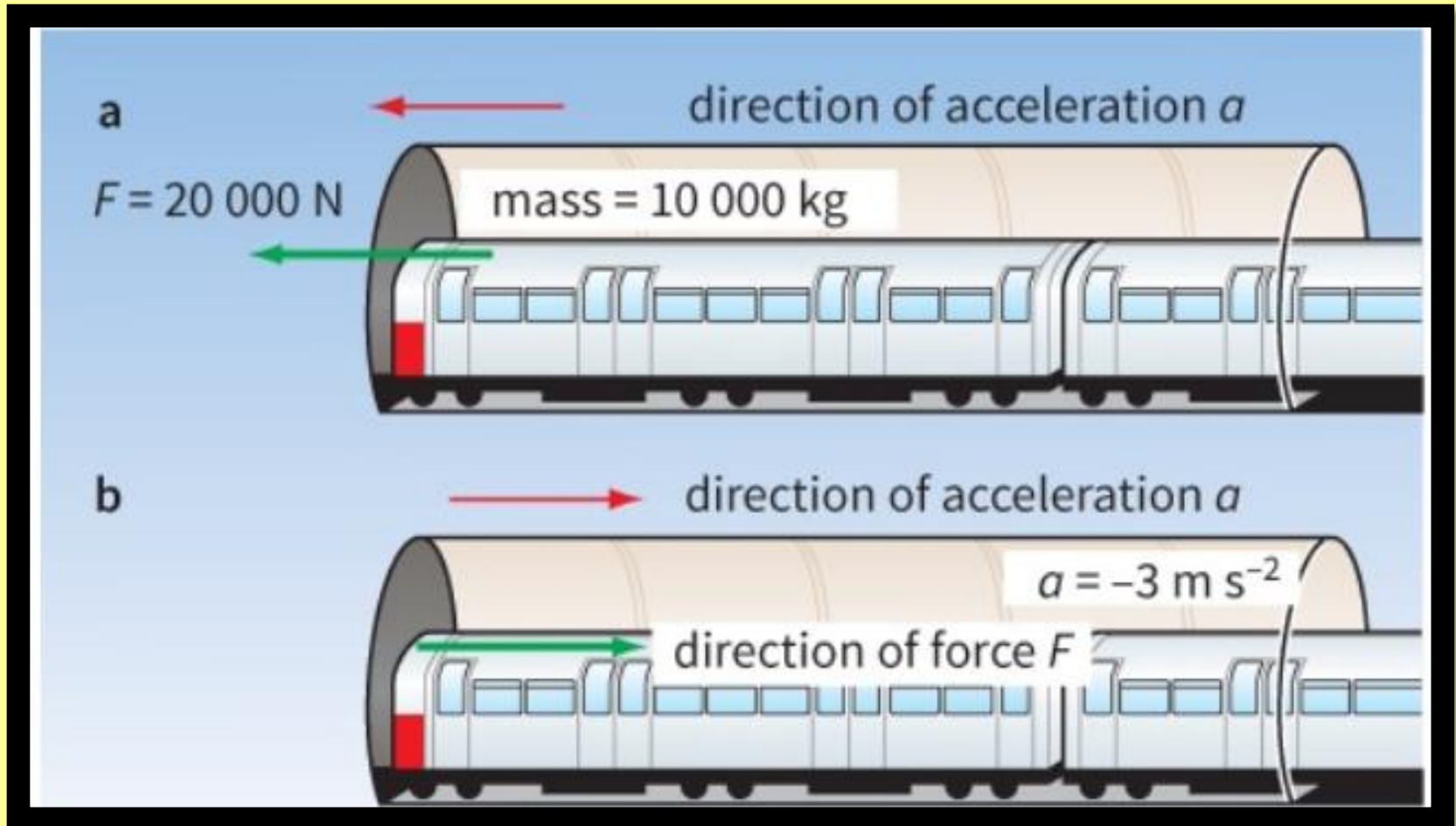
### Lab

Describe an experimental demonstration of using the Pressure sensor

# Dynamics: Explaining Motion



# Force, mass and acceleration



**Figure 3.2:** A force is needed to make the train **a** accelerate, and **b** decelerate.

# Newton's second law of motion

## KEY EQUATION

resultant force = mass  $\times$  acceleration

$$F = ma$$

In this example, we have  $F = 20\,000\text{ N}$  and  $m = 10\,000\text{ kg}$ , and so:

$$a = \frac{F}{m} = \frac{20\,000}{10\,000} = 2\text{ m s}^{-2}$$

In Figure 3.2b, the train is decelerating as it comes into a station. Its acceleration is  $-3.0\text{ m s}^{-2}$ . What force must be provided by the braking system of the train?

$$F = ma = 10\,000 \times -3 = -30\,000\text{ N}$$

The minus sign shows that the force must act towards the right in the diagram, in the opposite direction to the motion of the train.

# Example 1

- 1** A cyclist of mass 60 kg rides a bicycle of mass 20 kg. When starting off, the cyclist provides a force of 200 N. Calculate the initial acceleration.

**Step 1** This is a straightforward example. First, we must calculate the combined mass  $m$  of the bicycle and its rider:

$$m = 20 + 60 = 80 \text{ kg}$$

We are given the force  $F$ :

force causing acceleration  $F = 200 \text{ N}$

**Step 2** Substituting these values gives:

$$\begin{aligned} a &= \frac{F}{m} \\ &= \frac{200}{80} \\ &= 2.5 \text{ m s}^{-2} \end{aligned}$$

So the cyclist's acceleration is  $2.5 \text{ m s}^{-2}$ .

# Example 2

**2** A car of mass 500 kg is travelling at  $20 \text{ m s}^{-1}$ . The driver sees a red traffic light ahead, and slows to a halt in 10 s. Calculate the braking force provided by the car.

**Step 1** In this example, we must first calculate the acceleration required. The car's final velocity is  $0 \text{ m s}^{-1}$ , so its change in velocity  $\Delta v = 0 - 20 = -20 \text{ m s}^{-1}$

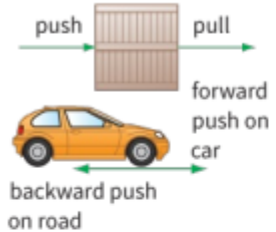
$$\begin{aligned}\text{acceleration } a &= \frac{\text{change in velocity}}{\text{time taken}} \\ &= \frac{\Delta v}{\Delta t} \\ &= \frac{-20}{10} \\ &= -2 \text{ m s}^{-2}\end{aligned}$$

**Step 2** To calculate the force, we use:

$$F = ma = 500 \times -2 = -1000 \text{ N}$$

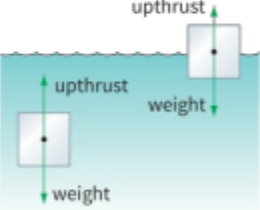
So the brakes must provide a force of 1000 N. (The minus sign shows a force decreasing the velocity of the car.)

# Identifying forces

| Diagram                                                                                                                                                           | Force                                                                                                                                                                                                                                                                                                                                                                                                                     | Important situations                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>push</p> <p>pull</p> <p>forward push on car</p> <p>backward push on road</p> | <p><b>Pushes and pulls.</b> You can make an object accelerate by pushing and pulling it. Your force is shown by an arrow pushing (or pulling) the object.</p> <p>The engine of a car provides a force to push backwards on the road. Frictional forces from the road on the tyre push the car forwards.</p>                                                                                                               | <ul style="list-style-type: none"> <li>pushing and pulling</li> <li>lifting</li> <li>force of car engine</li> <li>attraction and repulsion by magnets and by electric charges</li> </ul> |
|  <p>weight</p>                                                                   | <p><b>Weight.</b> This is the force of gravity acting on the object. It is usually shown by an arrow pointing vertically downwards from the object's centre of gravity.</p>                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>any object in a gravitational field</li> <li>less on the Moon</li> </ul>                                                                          |
|  <p>friction</p> <p>pull</p> <p>friction</p>                                    | <p><b>Friction.</b> This is the force that arises when two surfaces rub over one another. If an object is sliding along the ground, friction acts in the opposite direction to its motion. If an object is stationary, but tending to slide - perhaps because it is on a slope - the force of friction acts up the slope to stop it from sliding down. Friction always acts along a surface, never at an angle to it.</p> | <ul style="list-style-type: none"> <li>pulling an object along the ground</li> <li>vehicles cornering or skidding</li> <li>sliding down a slope</li> </ul>                               |



# Identifying forces

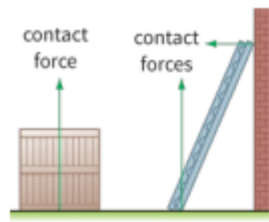
|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>Drag.</b> This force is similar to friction. When an object moves through air, there is friction between it and the air. Also, the object has to push aside the air as it moves along. Together, these effects make up drag.</p> <p>Similarly, when an object moves through a liquid, it experiences a drag force.</p> <p>Drag acts to oppose the motion of an object; it acts in the opposite direction to the object's velocity. It can be reduced by giving the object a streamlined shape.</p>    | <ul style="list-style-type: none"> <li>● vehicles moving</li> <li>● aircraft flying</li> <li>● parachuting</li> <li>● objects falling through air or water</li> <li>● ships sailing</li> </ul> |
|  | <p><b>Upthrust.</b> Any object placed in a fluid such as water or air experiences an upwards force. This is what makes it possible for something to float in water.</p> <p>Upthrust arises from the pressure that a fluid exerts on an object. The deeper you go, the greater the pressure. So there is more pressure on the lower surface of an object than on the upper surface, and this tends to push it upwards. If upthrust is greater than the object's weight, it will float up to the surface.</p> | <ul style="list-style-type: none"> <li>● boats and icebergs floating</li> <li>● people swimming</li> <li>● divers surfacing</li> <li>● a hot air balloon rising</li> </ul>                     |

## KEY EQUATION

$$\text{weight} = \text{mass} \times \text{acceleration of free fall}$$

$$W = mg$$

# Identifying forces

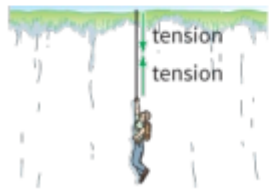


**Contact force.** When you stand on the floor or sit on a chair, there is usually a force that pushes up against your weight, and which supports you so that you do not fall down. The contact force is sometimes known as the normal contact force of the floor or chair. (In this context, normal means 'perpendicular'.)

- standing on the ground
- one object sitting on top of another
- leaning

The contact force always acts at right angles to the surface that produces it. The floor pushes straight upwards; if you lean against a wall, it pushes back against you horizontally.

- against a wall
- one object bouncing off another

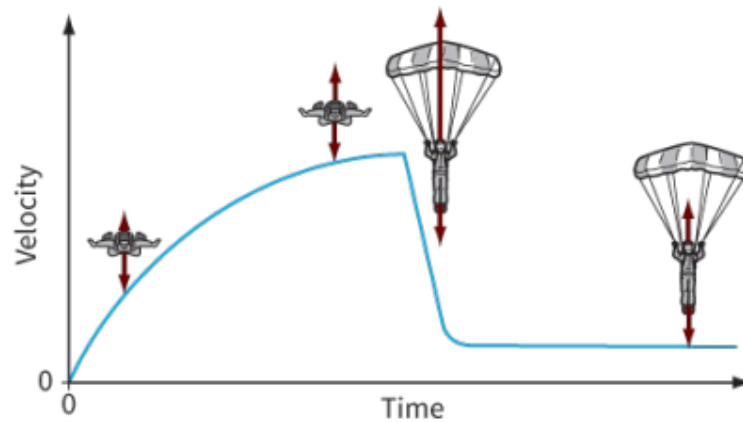


**Tension.** This is the force in a rope or string when it is stretched. If you pull on the ends of a string, it tends to stretch. The tension in the string pulls back against you. It tries to shorten the string.

Tension can also act in springs. If you stretch a spring, the tension pulls back to try to shorten the spring. If you squash (compress) the spring, the tension acts to expand the spring.

- pulling with a rope
- squashing or stretching a spring

# What is Terminal Velocity and how it is related to force



# Moving through air

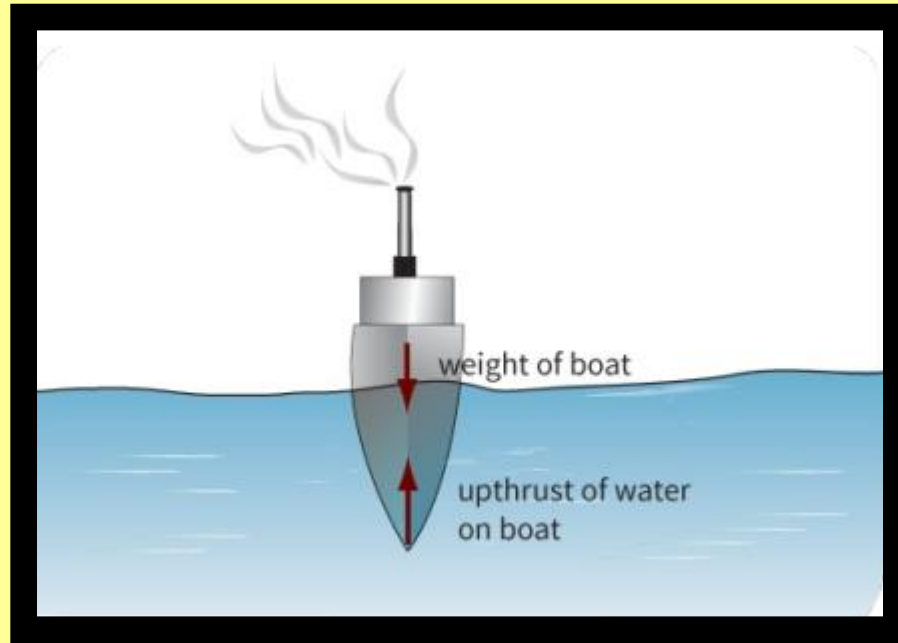


A racing cyclist adopts a posture that helps to reduce drag. Clothing, helmet and even the cycle itself are designed to allow them to go as fast as possible.



A runner making use of air resistance to build up his muscles.

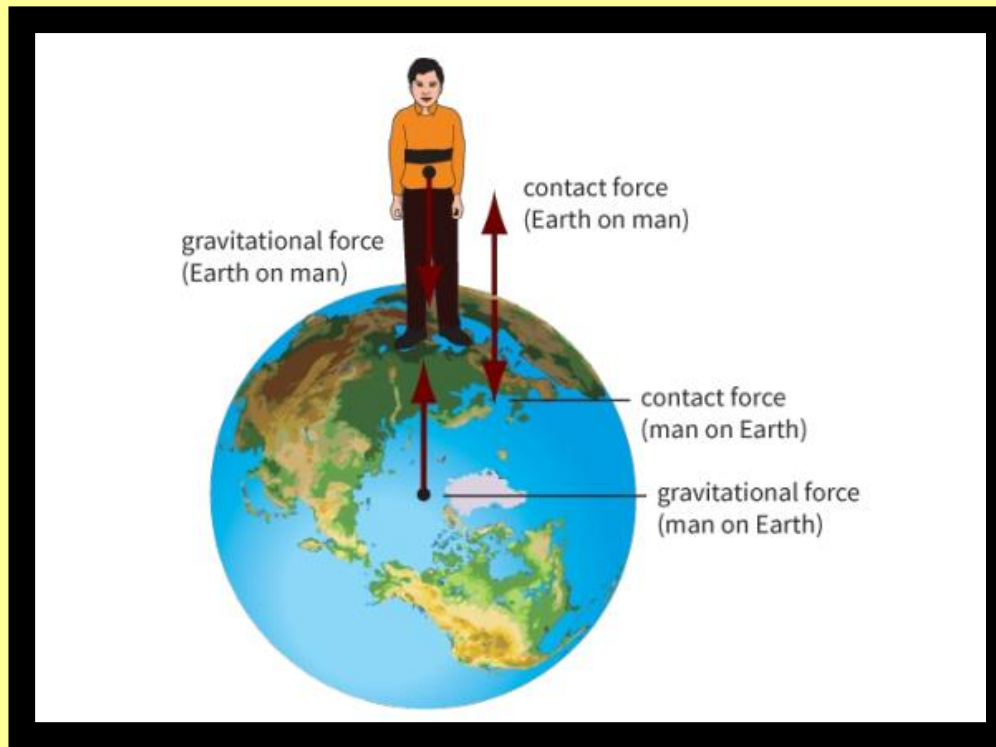
# Moving through Liquid



**Without sufficient upthrust from the water, the boat would sink**

# Newton's third law of motion

- They act on different objects.
- They are equal in magnitude.
- They are opposite in direction.
- They are forces of the same type.



**For each of the forces that the Earth exerts on you, an equal and opposite force acts on the Earth.**

# The seven SI base units

| Base unit                 | Symbol              | Base unit     |
|---------------------------|---------------------|---------------|
| length                    | $x, l, s$ and so on | m (metre)     |
| mass                      | $m$                 | kg (kilogram) |
| time                      | $t$                 | s (second)    |
| electric current          | $I$                 | A (ampere)    |
| thermodynamic temperature | $T$                 | K (kelvin)    |
| amount of substance       | $n$                 | mol (mole)    |
| luminous intensity        | $I$                 | cd (candela)  |



## SUMMARY

An object will remain at rest or in a state of uniform motion unless it is acted on by an external force. This is Newton's first law of motion.

For a body of constant mass, the acceleration is directly proportional to the resultant force applied to it. Resultant force  $F$ , mass  $m$  and acceleration  $a$  are related by the equation:

$$\text{resultant force} = \text{mass} \times \text{acceleration} (F = ma)$$

This is a form of Newton's second law of motion.

When two bodies interact, the forces they exert on each other are equal in magnitude and opposite in direction. This is Newton's third law of motion.

The acceleration produced by a force is in the same direction as the force. Where there are two or more forces, we must determine the resultant force.

A newton (N) is the force required to give a mass of 1 kg an acceleration of  $1 \text{ m s}^{-2}$  in the direction of the force.

The greater the mass of an object, the more it resists changes in its motion. Mass is a measure of the object's inertia.

The weight of an object is a result of the pull of gravity on it:

$$\text{weight} = \text{mass} \times \text{acceleration of free fall} (W = mg)$$

Terminal velocity is reached when the fluid resistance is equal to the weight of the object.

Physics equations are homogenous and have the same base units on each side. The main base units are m, kg, s, A and K (the thermodynamic unit for temperature).



الْعِلْمُ مَا نَفَعَهُ وَ لَيْسَ مَا حَفِظَ

Science is what should be applied, not  
just memorised